AtiNAMatiC-Gate -- User Manual

Version 0.1.0 · Atij 666 Studio · Windows (Standalone, VST3 & CLAP)

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1. What it is

AtiNAMatiC-Gate is a **professional noise gate** built specifically for high-gain guitar. One large knob controls everything -- from threshold and hysteresis to hold time and aggressiveness -- so you stay focused on playing instead of tweaking.

Under the hood it is a full pro-grade gate engine:

- **Sidechain HPF at 90 Hz** -- low-end rumble and hum cannot false-trigger the gate.
- **Dual-time-constant detector** -- a fast peak follower catches transients; a slow RMS follower tracks sustain. Both feed the same decision circuit.
- **Hysteresis (open/close gap)** -- the close threshold is always lower than the open threshold, so the gate never chatters at the boundary.
- **Adaptive noise-floor tracking** -- in Adaptive mode the open threshold automatically lifts 10 dB above the measured noise floor, so a warming amp stays gated even as it heats up.
- **Hold time** -- after the signal falls below the close threshold the gate stays fully open for a short hold period, eliminating end-of-note chatter on palm mutes.
- **Asymmetric attack/release with cubic easing** -- smooth, click-free transitions that do not add audible artifacts.
- **3 ms audio lookahead** -- the gate opens *before* the transient arrives, so it never chops a pick attack even at the Tightest setting.
- **Soft attenuation floor (-65 dB)** -- the gate attenuates instead of hard-muting, so bleed and amp hiss fade out rather than snapping off.

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2. Signal chain placement

```
Guitar  -->  Amp Sim (NAM / Helix / ML Sound Lab / etc.)  -->  Cab / IR
          -->  *** AtiNAMatiC-Gate ***  -->  AtiNAMatiC-Air / Reverb / Master
```

Place the gate **directly after your amp sim / cab IR**, before any time-based effects. Gating after reverb or delay cuts the tail; gating before the amp sim may change the amp's natural compression and feel.

For a parallel rig (wet/dry/wet), insert the gate on each dry cab channel individually.

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3. Controls

GATE (large centre knob, 0 -- 100%)

The single master macro. As you turn it clockwise:

GATE value	Open threshold	Hysteresis gap	Hold time	Aggressiveness
0% (OPEN)	-78 dB	4 dB	longest	gentlest
50%	-50 dB	6 dB	medium	moderate

100% (TIGHT)	-22 dB	8 dB	shortest	tightest

Start at 35--50% for most high-gain situations. Increase until the noise between notes disappears. Decrease if the gate is chopping pick attacks or cutting note tails.

The current value is shown as a percentage in the label directly below the knob. Tick marks label the three key positions: **OPEN**, **TRACKING**, and **TIGHT**.

MODE (three-position selector)

Chooses the time constants used by the gate engine:

Mode	Attack	Release	Hold base	Best for
Adaptive (default)	1.5 ms	90 ms	60 ms	Most situations. Threshold follows the noise floor automatically.
Fast	0.4 ms	25 ms	20 ms	Tight metal / djent riffs where you need near-instant gating between fast palm-mute chops.
Natural	2.5 ms	260 ms	120 ms	Clean or lightly driven tones, leads, and any source where you want the gate to breathe.

Hold time scales down with the GATE knob in all modes: the tighter the gate, the shorter the hold.

MIX (0 -- 100%, default 100%)

Dry/wet blend. At 100% only the gated signal is heard. Lower MIX blends in the original dry signal in parallel, useful if you want just a hint of noise reduction without touching the note tails. Default 100%.

OUTPUT (-24 to +12 dB, default 0 dB)

Master output level after gating. Use it to compensate for level differences between gate-open and gate-closed moments in a mix, or simply to match the level of an adjacent track. The value is displayed inside the knob's text box.

ACTIVITY meter (centre panel)

Two horizontal bars showing real-time signal status:

Bar	What it shows
INPUT	Peak level of the signal entering the gate (after the lookahead buffer)
THRESHOLD	The current open threshold -- moves in Adaptive mode as the noise floor is tracked

The dB scale below the meter runs from -60 dB to 0 dB.

State pill (bottom centre, below the meter)

A three-state indicator showing the gate's current condition:

Pill label	Meaning
CLOSED	Gate is attenuating (-65 dB floor applied)

OPEN	Gate is fully open, passing signal at unity
(transitioning)	Gate is in attack or release phase (pill fades between states)

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4. Quick-start settings

General high-gain rhythm

- MODE: Adaptive -- GATE: 45% -- MIX: 100% -- OUTPUT: 0 dB

Tight djent / metal (fast riffs)

- MODE: Fast -- GATE: 65% -- MIX: 100% -- OUTPUT: 0 dB

Noisy vintage amp (hum reduction)

- MODE: Adaptive -- GATE: 30% -- MIX: 100% -- OUTPUT: 0 dB *(Low GATE keeps threshold below the amp's harmonic floor so sustain is not affected.)*

Lead / single-note lines

- MODE: Natural -- GATE: 35% -- MIX: 100% -- OUTPUT: 0 dB

Parallel blend (keep natural tails)

- MODE: Adaptive -- GATE: 60% -- MIX: 60% -- OUTPUT: 0 dB *(Gated signal blended with 40% dry preserves the natural decay while cutting most noise.)*

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5. Latency

AtiNAMatiC-Gate introduces exactly **3 ms of lookahead latency** at any sample rate.

This latency is **reported to your DAW** via the standard plug-in latency API, so PDC (Plugin Delay Compensation) is applied automatically -- your gated channel stays phase-aligned with every other track with no manual offset required.

At 44.1 kHz the lookahead is 132 samples. At 48 kHz it is 144 samples. At 96 kHz it is 288 samples.

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6. System requirements

	Windows	macOS	Linux
OS	Windows 10 / 11 (64-bit)	macOS 10.15 Catalina or later	Ubuntu 20.04+ / equivalent
Formats	Standalone, VST3, CLAP	Standalone, VST3, AU, CLAP	Standalone, VST3, CLAP
CPU	x86-64	Apple Silicon (arm64) or Intel x86-64	x86-64
RAM	4 GB minimum	4 GB minimum	4 GB minimum
Sample rates	44.1 -- 192 kHz	44.1 -- 192 kHz	44.1 -- 192 kHz

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7. Installation

Windows -- using the installer

1. Run **AtiNAMatiC-Gate-Setup-0.1.0.exe** as Administrator.
2. Select the formats to install (Standalone is always included):
   - **VST3 plug-in** installs to `C:\Program Files\Common Files\VST3\AtiNAMatiC-Gate.vst3`
   - **CLAP plug-in** installs to `C:\Program Files\Common Files\CLAP\AtiNAMatiC-Gate.clap`
3. Click Install. When complete, a dialog will remind you to rescan plug-ins in your DAW.
4. Rescan (see table below).

Windows -- manual installation

Copy the files to their locations manually:

File	Destination
`AtiNAMatiC-Gate.exe`	Anywhere you like (e.g. `C:\Program Files\Atij 666 Studio\AtiNAMatiC-Gate\`)
`AtiNAMatiC-Gate.vst3` (folder)	`C:\Program Files\Common Files\VST3\`
`AtiNAMatiC-Gate.clap`	`C:\Program Files\Common Files\CLAP\`

macOS -- manual installation

File	Destination
`AtiNAMatiC-Gate.vst3`	`/Library/Audio/Plug-Ins/VST3/`
`AtiNAMatiC-Gate.component` (AU)	`/Library/Audio/Plug-Ins/Components/`
`AtiNAMatiC-Gate.clap`	`/Library/Audio/Plug-Ins/CLAP/`

On Apple Silicon, open **Audio MIDI Setup** to confirm the plug-in shows as arm64 (not Rosetta).

Linux -- manual installation

File	Destination
`AtiNAMatiC-Gate.vst3` (folder)	`~/.vst3/` or `/usr/lib/vst3/`
`AtiNAMatiC-Gate.clap`	`~/.clap/` or `/usr/lib/clap/`

DAW rescan

DAW	Rescan command
Reaper	Options > Preferences > Plug-ins > VST > Re-scan
Cubase / Nuendo	Studio > VST Plug-in Manager > Update
Studio One	Options > Locations > Plug-ins > Rescan
Ableton Live	Preferences > Plug-Ins > Rescan

Bitwig Studio	Settings > Plug-ins > Rescan
FL Studio	Options > Manage plug-ins > Start scan
Logic Pro (AU)	Audio MIDI Setup -- plug-ins validate automatically on launch

The plug-in appears under manufacturer **Atij666Studio**, category **Dynamics / Gate**.

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8. Controls reference

Control	Range	Default	Description
GATE	0 -- 100%	35%	Master macro. Controls threshold (-78 to -22 dB), hysteresis (4--8 dB), hold time, and aggressiveness.
MODE	Adaptive / Fast / Natural	Adaptive	Selects the time constants for attack, release, and hold.
MIX	0 -- 100%	100%	Dry/wet parallel blend.
OUTPUT	-24 to +12 dB	0 dB	Master output level after gating.

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9. Internal signal chain

```
Input
  |
  +-- [Sidechain copy] --> HPF @ 90 Hz --> Peak follower + RMS detector
  |                                                |
  |                              Adaptive noise floor tracker
  |                                                |
  |                              Gate decision (open / hold / close)
  |                                                |
  |                              Attack / release with cubic easing
  |                                                |
  +-- [3 ms lookahead delay] ----------> Gain apply (floor: -65 dB)
                                                   |
                                          MIX (parallel blend with dry)
                                                   |
                                          OUTPUT gain
                                                   |
                                            Output
```

The sidechain path is always mono (left channel only for stereo signals) so the gate decision is consistent regardless of stereo phase differences. The gain is applied identically to all output channels.

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10. Typical use cases

Rhythm guitar (high gain) Insert after your amp sim / cab IR. Set MODE: Adaptive, GATE: 45%. The gate silences hum and hiss between riffs while the 3 ms lookahead ensures no pick attack is ever lost. Hold time keeps notes from choking on faster riffs.

Lead guitar Set MODE: Natural, GATE: 30--40%. The longer release (260 ms) preserves sustain and lets vibrato notes decay naturally. MIX at 100%. Lower GATE if the gate is audibly squeezing the end of long bends.

Drop-tuned / djent (very fast riffs) Set MODE: Fast, GATE: 60--75%. The 0.4 ms attack is near-instant. If the gate is still letting a tiny burst of pre-note hiss through, increase GATE by 5--10%.

Vintage amp hum Set MODE: Adaptive, GATE: 25--35%. Adaptive mode tracks the hum floor and parks the threshold just above it, so the gate closes as tightly as needed without needing manual threshold adjustment when the amp warms up.

Parallel noise reduction Set MIX: 40--70%. This blends gated and dry signals together. The noise is reduced without completely removing the natural tail of each note. Works well on clean or lightly crunch tones where hard gating sounds unnatural.

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11. Tips

- **If pick attacks are still being clipped:** lower GATE toward 30%. The threshold may be too high, closing the gate before the pick fully arrives in the lookahead window. Alternatively, check that the gate is after the amp sim (not before) -- amp sims compress the transient which can confuse the detector.
- **If hiss bleeds through between notes:** raise GATE toward 60%. Also check MODE is not set to Natural (long release lets more noise through at the tail of each note).
- **Use GATE + MIX together for parallel gating:** GATE at 70%, MIX at 50% is a common starting point. The gated copy removes most noise while the 50% dry blend preserves natural feel.
- **Adaptive mode with a cold amp:** if your amp sim's noise floor changes as the session warms up, Adaptive tracks it automatically. You rarely need to touch the GATE knob between takes.
- **Automation:** all four parameters (GATE, MODE, MIX, OUTPUT) are fully automatable in any DAW. MODE changes are per-block, so automate it only at section boundaries to avoid audible clicks.
- **3 ms latency:** at 44.1 kHz this is 132 samples -- easily absorbed by any modern DAW with PDC enabled. If you are routing to external hardware, offset your hardware outputs by the same amount.

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12. About

- Design & development: **Atij** (Atij 666 Studio)
- Built with **JUCE 8** (juce.com)
- DSP inspired by the professional gate designs found in classic hardware units
- Neural Amp Modeler (NAM) by **Steven Atkinson** (the amp sims this gate was built to complement)
- Thank you to the open-source audio and guitar-plugin community

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13. License & support

Personal and commercial use permitted on machines you own. Bug reports and feature requests: click the **AtiNAMatiC-Gate** title badge in the plug-in window to open the About dialog.

AtiNAMatiC-Gate is part of the AtiNAMatiC Series -- a family of focused, hardware-flavoured guitar tools for the modern post-amp signal chain.